

1 The radial wavefunction

We will closely follow the treatment of hydrogen in the Feynman lectures on physics. We first begin with a simplified model in which we neglect angular momentum and spin of the electron and only consider its radial position. The schrodinger equation reads:

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$$

where ψ is the wavefunction and we take $V = -\frac{e^2}{4\pi\epsilon_o}$ caused by a proton at the origin, m is the electron mass. In cartesian co-ordinates the vector operator $\vec{\nabla}$ (where we wrote the Laplacian $\nabla^2 = \vec{\nabla} \cdot \vec{\nabla}$) has a simple form:

$$\vec{\nabla} = (\partial_x, \partial_y, \partial_z)$$

Since we want to describe the electron's distance from the origin (the proton) it is clearly best to pick spherical co-ordinates given by:

$$x = r \sin(\theta) \cos(\phi)$$

$$y = r \sin(\theta) \sin(\phi)$$

$$z = r \cos(\theta)$$

We quote that the radial part of the Laplacian in spherical coordinates is:

$$\nabla^2 f = \frac{1}{r} \partial_r^2 (rf)$$

We then have:

$$-\frac{\hbar^2}{2m} \frac{1}{r} \partial_r^2 (r\psi) - \frac{e^2}{4\pi\epsilon_o} \psi = E\psi$$

To clean the equation up we choose units so $4\pi\epsilon_o = 1$ if we define $r = \hbar^2 \rho / me^2$ and use the potential $\varepsilon = 2\hbar^2 E / me^4$ then we have:

$$\frac{1}{\rho} \frac{d^2}{d\rho^2} \rho\psi \frac{m^2 e^4}{\hbar^4} = -\frac{2m}{\hbar^2} \left(\frac{me^4}{\hbar^2} \varepsilon + \frac{me^4}{\hbar^2 \rho} \right)$$

$$\frac{1}{\rho} \frac{d^2}{d\rho^2} \rho\psi = -\left(\varepsilon + \frac{2}{\rho} \right)$$

$r_B = \hbar^2/me^2$ and $E_R = e^2m/2\hbar^2$ are the 'Bohr radius' and 'Rydberg constant' and are convenient to define because, as we will see, the hydrogen atom will have discrete energy levels. Feynman says "There are explanations [of how to solve this DE] ... but they are not real." Thus we will just show one method below while keeping in mind some steps may seem arbitrary but are used because they work.

let $f = \rho\psi$ and $g(\rho)$ so that $f(\rho) = e^{-\alpha\rho}g(\rho)$

$$g' = e^{\alpha\rho}[fa + f']$$

so

$$g'' = e^{\alpha\rho}[fa^2 + 2f'a + f'']$$

putting this back into schrodinger's equation:

$$g'' - 2\alpha\rho g' + \left(\frac{2}{\rho} + \alpha^2 + \varepsilon\right)g$$

choosing $\alpha^2 = -\varepsilon$:

$$g'' - 2\alpha\rho g' + \frac{2}{\rho}g$$

We then expand g in an infinite power series of ρ :

$$g = \sum b_k k \rho^{k-1}$$

Note that from the power law for differentiation:

$$g' = \sum b_k \rho^{k-1} k$$

$$g'' = \sum b_{k+1} (k+1) \rho^{k-1}$$

So the schrodinger equation reads:

$$\rho^{k-1}(b_{k+1}(k+1) - 2\alpha b_k k + 2b_k) = 0$$

Clearly this only holds for all ρ if the second bracket is zero. But this relates b_{k+1} to b_k :

$$b_{k+1} = 2b_k \frac{\alpha k - 1}{k(k+1)}$$

In order to find the energy levels we could sub back our solution into the schrodinger equation. A nicer argument (which does not involve any tedious

differentiation) is considering when the electron is far from the proton and demanding the wavefunction goes to zero (so the electron is bound.) For large k we have:

$$\frac{\alpha k - 1}{k + 1} \approx \alpha$$

so

$$b_{k+1} \approx \frac{2\alpha b_k}{k} \approx \frac{(2\alpha)^k}{k!}$$

recall that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \dots$$

so that our coefficients give:

$$g(\rho) \approx e^{2\alpha\rho}$$

$$f(\rho) \approx e^{\alpha\rho}$$

but this appears to be (exponentially) divergent as ρ gets large. The only way the series terminates is if $\alpha = 1/n$ where n is any integer, so that terms for $k > n$ vanish since:

$$b_{n+1} = 2b_n \frac{\frac{n}{n} - 1}{n(n+1)} = 0$$

Recalling that $\alpha^2 = -\varepsilon$ and that ε was in Rydberg constant units we have in normal units

$$E_n = -\frac{E_R}{n^2} = -\frac{me^4}{32\pi^2\varepsilon_0\hbar^2n^2}$$

These are the allowed energies that just happen to be exactly equal to those Bohr derived using a mix of classical and quantum physics thirteen years before Schrodinger did using his equation like us!

2 Emission spectra

That was a long detour into some pretty heavy quantum mechanics, although it was rewarding to see how Bohr's formula comes from the Schrodinger equation itself and not have to rely on arguments with classical orbitals. Now we can quite easily find the wavelength of emitted radiation. If an electron moves from energy level m to n then the change in energy is equal to the energy gained by the emitted photon. Using Planck's formula for the energy of a photon:

$$\Delta E = E_m - E_n = \hbar\omega$$

and recalling the wave equation:

$$\begin{aligned}\lambda &= c/f = 2\pi c/\omega = \frac{2\pi c\hbar}{\Delta E} \\ &= \frac{2\pi\hbar c}{E_R} \left(\frac{1}{n^2} - \frac{1}{m^2} \right)^{-1}\end{aligned}$$

Of course actual hydrogen emission spectra differ slightly: 'fine structure' emerges from spin-orbit coupling, there is a relativistic correction and even smaller differences due to the proton itself being a magnetic dipole or the nucleus being a finite size (not a point mass and with constant potential inside.)